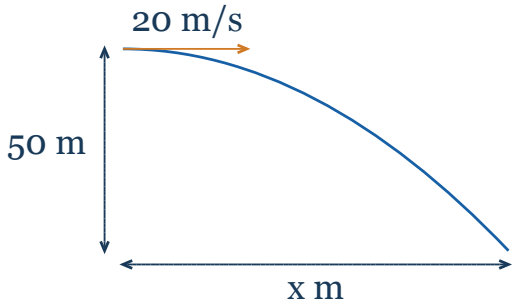
Projectiles

Overview

In Mechanics Year 1, vertical projection used the suvat equations with acceleration $g\,\mathrm{m\,s^{-2}}$ downwards. Here objects can also be projected sideways.

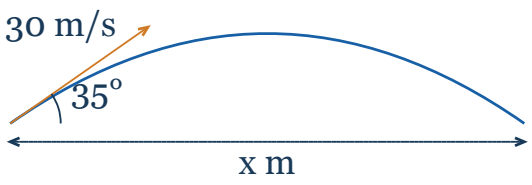
1: Horizontally projected

A particle is projected horizontally at $20\,\mathrm{m\,s^{-1}}$, 50 m above the ground. How far along the ground does it travel?



2: Projection at any angle

A cabbage is projected from ground level at $30\,\mathrm{m\,s^{-1}}$, at 35° to the horizontal. How far away does it hit the ground?



Acceleration in Each Direction 3

Consider vertical and horizontal motion separately.

Vertical acceleration downwards:

$g\,\mathrm{m\,s^{-2}}$. Use the suvat equations.

Horizontal acceleration:

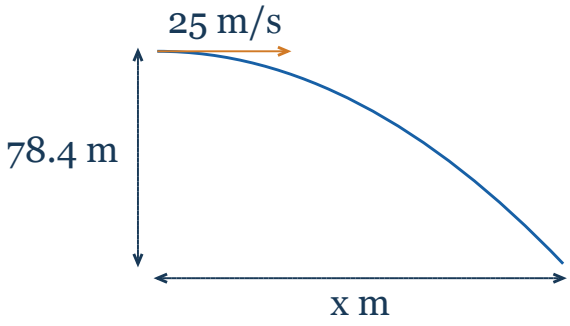
$0\,\mathrm{m\,s^{-2}}$. Velocity is constant, so speed = distance/time.

A particle is projected horizontally at $25\,\mathrm{m\,s^{-1}}$ from 78.4 m above a horizontal surface. Find:

a. The time to reach the surface.

b. The horizontal distance travelled.

c. The distance of the impact point from the original point.



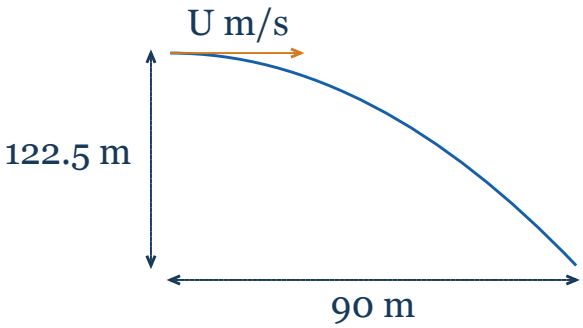
a

b

c

Further Example 4

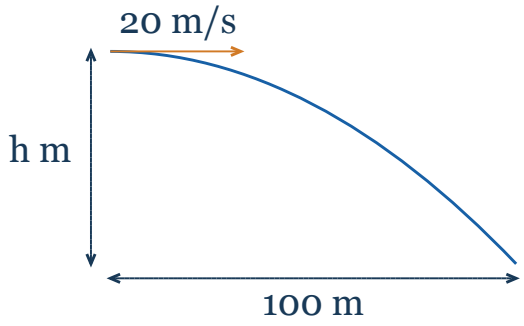
A particle is projected horizontally at $U\,\mathrm{m\,s^{-1}}$ from 122.5 m above a horizontal plane. It lands 90 m horizontally from its starting point. Find its initial speed.



Working

Test Your Understanding 5

A particle is projected horizontally at $20\,\text{m s}^{-1}$ from h m above a horizontal plane. It lands 100 m horizontally from its starting point. Determine h.



Working

Exercise 6A — Y2 Mechanics Pages 110–111 6

5. A particle moves horizontally at $20\,\text{m s}^{-1}$ along a flat smooth table, starting 2 m from its edge. The table is 1.2 m above the floor. Find the total time from projection until it lands on the floor.

Working

Working

6. A dart is thrown horizontally at $14\,\text{m s}^{-1}$ towards a vertical dartboard. Model the dart as a particle moving freely under gravity, in a vertical plane perpendicular to the board. It hits 9 cm below its launch level. Find the horizontal distance from the launch point to the board.

Working

Working

7. A particle of mass 2.5 kg is projected along a horizontal rough surface at $5\,\text{m s}^{-1}$. After 2 m, it leaves as a projectile and lands 1.2 m below. Total time from projection to landing is 1.0 s. Find:

a. The time in contact with the surface.

b. The coefficient of friction between particle and surface.

a

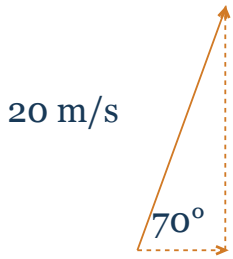
a

b

b

Components of Velocity 9

As with forces, split velocity into horizontal and vertical components.

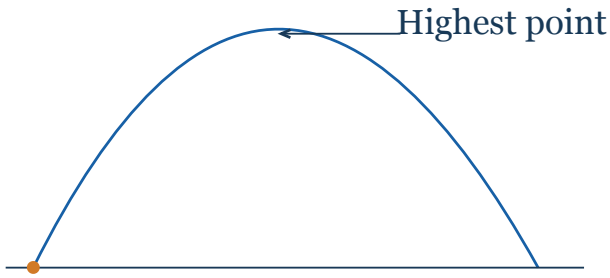


Horizontal component:

$20\cos 70^\circ \text{ m s}^{-1}$.

Vertical component:

$20\sin 70^\circ \text{ m s}^{-1}$.



At the highest point, the vertical velocity is:

0.

Speed is the magnitude of velocity. Initially:

$\mathbf{v} \approx \begin{pmatrix} 6.84 \\ 18.8 \end{pmatrix} \text{ m s}^{-1}$.

$$|\mathbf{v}| = \sqrt{(20\cos 70^\circ)^2 + (20\sin 70^\circ)^2}$$
$$\approx \sqrt{6.84^2 + 18.8^2} \approx 20 \text{ m s}^{-1}.$$

Simple Example 10

A particle P is projected from O on a horizontal plane at 28 m s^{-1}, at elevation 30°. It moves freely under gravity until it strikes the plane at A. Find:

a. Its greatest height above the plane.

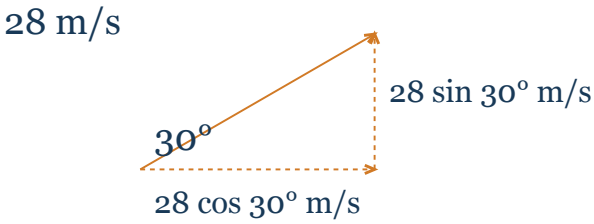
b. Its time of flight.

c. The distance OA.

a

b

c



Projected from Above Ground 11

A particle is projected from O at $V\ \mathrm{m\,s^{-1}}$, at elevation θ, where $\tan\theta = 4/3$. Point O is $42.5\ \mathrm{m}$ above a horizontal plane. The particle strikes the plane at A after $5\ \mathrm{s}$.

a. Show that $V = 20\ \mathrm{m\,s^{-1}}$.

b. Find OA.

Draw a diagram before solving.

Diagram

a

a

b — horizontal

b — distance

Time Above a Given Point 12

A particle is projected from O at $35\ \mathrm{m\,s^{-1}}$, at elevation $30°$. It moves freely under gravity. Find the length of time for which it is $15\ \mathrm{m}$ or more above O.

Find the two times at height $15\ \mathrm{m}$; subtract them.

Working

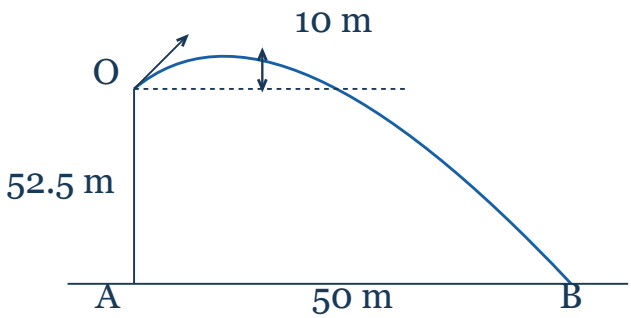
Edexcel M2 (old), May 2012, Q7

A small stone is projected from O, at the top of a vertical cliff OA, 52.5 m above the sea. It rises 10 m above O, then hits the sea at B, with $AB = 50$ m. Model it as a particle moving freely under gravity.

a. Show its initial vertical velocity component is $14\,\mathrm{m\,s^{-1}}$. (3 marks)

b. Find its speed of projection. (9 marks)

c. Find the time when it moves parallel to OB. (5 marks)



a

b

c

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

7. A particle is projected with initial velocity $\mathbf{U} = 3k\mathbf{i} + 2k\mathbf{j}\,\mathrm{m\,s^{-1}}$.

a. Find the angle of projection.

b. Given its initial speed is $3\sqrt{13}\,\mathrm{m\,s^{-1}}$, find k.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

b

a

9. A stone is thrown at $30\,\text{m s}^{-1}$ from a window $20\,\text{m}$ above horizontal ground. It hits the ground $3.5\,\text{s}$ later. Find:

a. Its angle of projection.

b. The horizontal distance from the window to its landing point.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

a

b

10. A ball is thrown from O on horizontal ground at $U\,\text{m s}^{-1}$, at elevation θ, where $\tan\theta = 3/4$. It strikes a vertical wall $20\,\text{m}$ from O, $3\,\text{m}$ above the ground. Find:

a. U.

b. The time from projection to striking the wall.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

a

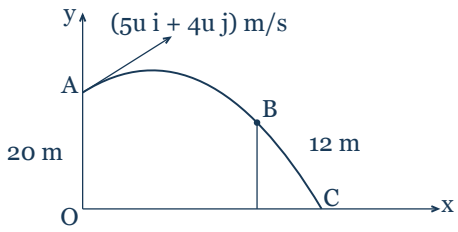
b

11. Particle P is projected from A, position $20\mathbf{j}$ m relative to fixed origin O, with velocity $(5u\mathbf{i} + 4u\mathbf{j})\,\mathrm{m\,s^{-1}}$. It moves freely under gravity through B, position $(k\mathbf{i} + 12\mathbf{j})$ m, before reaching C on the x-axis. Travel from A to B takes 4 s. Find:

a. u.

b. k.

c. The angle its velocity makes with the x-axis at C.



Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

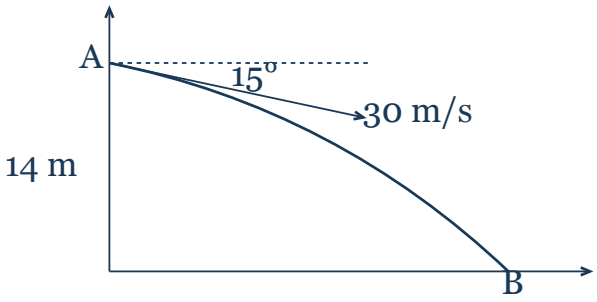
b

c

12. A stone is thrown from A at $30\,\mathrm{m\,s^{-1}}$, $15°$ below horizontal. Point A is 14 m above horizontal ground. It strikes the ground at B. Find:

a. The time from A to B.

b. The distance AB.



Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

b

13. A particle is projected from level ground at $U\,\mathrm{m\,s^{-1}}$, at elevation α. Its maximum height is $42\,\mathrm{m}$, and it lands $196\,\mathrm{m}$ from projection. Find α and U.

Working

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

Working

Working

15. An object is projected from fixed origin O at $(4\mathbf{i} + 5\mathbf{j})\,\mathrm{m\,s^{-1}}$. It moves freely under gravity through P, position $k(\mathbf{i} - \mathbf{j})\,\mathrm{m}$, where $k > 0$.

a. Find k.

b.i. Find its speed at P.

b.ii. Find its direction of motion at P.

a

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

b.i

b.ii

16. A basketball player stands on the floor 10 m from a basket at height 3.05 m. The ball is thrown from height 2 m, at 40° above horizontal. Model it as a particle moving in a vertical plane. It passes through the basket.

a. Find its initial speed.

b. State two factors ignored by modelling it as a particle.

a

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

b

Extension — Maximum Range 22

Extension

A ball is projected from level ground with fixed speed u_0, at elevation θ. Prove its horizontal range is greatest at $45°$.

Year 2 differentiation is required.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

Working

Projectile Formulae 23

You may be asked to prove general results using the same projectile methods.

A particle is projected from a horizontal plane at speed U, elevation α, and moves freely under gravity until it returns to the plane at B. Find expressions for:

a. The time of flight T.

b. The range R.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

b

Projectile Formulae — Trajectory 24

A particle is projected at speed U, elevation α, and moves freely under gravity. After horizontal displacement x, its height above launch is y.

a. Show that

$$y = x\tan\alpha - \frac{gx^2}{2U^2}(1 + \tan^2\alpha).$$

A particle is projected from O on a horizontal plane at $28\,\text{m s}^{-1}$. It passes through B, $32\,\text{m}$ horizontally from O and $8\,\text{m}$ above the plane.

b. Find both possible elevations α, to the nearest degree.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

Do not be intimidated by symbols: resolve horizontally and vertically as usual.

Projectile Formulae — Two Possible Angles 25

A particle is projected at speed U, elevation α, and moves freely under gravity. After horizontal displacement x, its height above launch is y.

a. Show that

$$y = x\tan\alpha - \frac{gx^2}{2U^2}(1+\tan^2\alpha).$$

A particle is projected from O on a horizontal plane at $28\,\text{m s}^{-1}$. It passes through B, $32\,\text{m}$ horizontally from O and $8\,\text{m}$ above the plane.

b. Find both possible elevations α, to the nearest degree.

b

b

b

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

General Results 26

Exam note: you may be asked to derive these results. Use the resolving and suvat techniques from this chapter rather than memorising formulae for solving problems.

For initial speed U, elevation α, motion freely under gravity, returning to the same horizontal plane:

Time of flight $= \dfrac{2U\sin\alpha}{g}$.

Time to greatest height $= \dfrac{U\sin\alpha}{g}$.

Range $= \dfrac{U^2\sin 2\alpha}{g}$.

Trajectory: $y = x\tan\alpha - \dfrac{gx^2}{2U^2}(1+\tan^2\alpha)$.

Here y is height above launch and x is horizontal displacement.

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

5. Use $g = 10\,\text{m s}^{-2}$ in this question. A particle is projected from O at speed U, elevation α, and moves freely under gravity. Its horizontal displacement is x, height above O is y.

a. Show that $y = x\tan\alpha - gx^2/(2U^2\cos^2\alpha)$.

A stone is thrown from P at the end of a pier, 15 m above sea level, at $8\,\text{m s}^{-1}$ and elevation $40°$. Model it as a particle moving freely under gravity.

b. Find its horizontal distance from P when 2 m above sea level.

a

b

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

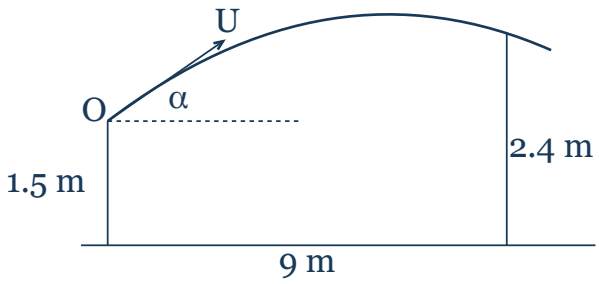
b

7. A volleyball player hits a ball towards a net 9 m away, from height 1.5 m. It moves in a vertical plane perpendicular to the net and just passes over its top, 2.4 m above ground. Model it as a particle projected from O at $U\,\text{m s}^{-1}$, elevation α.

a. Write expressions for horizontal and vertical displacements after t seconds and show at the net that

$$0.9 = 9\tan\alpha - \frac{81g}{2U^2\cos^2\alpha}.$$

b. Given $\alpha = 30°$, find the speed as it passes over the net.



Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

b

8. Unit vectors $\mathbf{i}$, $\mathbf{j}$ point horizontally and vertically upwards. An object is projected from A on horizontal ground at $(k\mathbf{i} + 2k\mathbf{j})\,\mathrm{m\,s^{-1}}$, $k > 0$. It moves freely under gravity until it lands at B and immediately stops. Take A as origin, with position $(x\mathbf{i} + y\mathbf{j})\,\mathrm{m}$.

a. Show that $y = 2x - gx^2/(2k^2)$.

b. Given range $AB = R$ and maximum height H, find in terms of k, g: i. R; ii. H.

a

Constant acceleration

$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$

$s = \frac{u+v}{2}t, \quad s = vt - \frac{1}{2}at^2.$

a

b.i

b.ii